

CIS022 day 8

more miscellany

Agenda

- A lil more on pass-by-reference
- quick style guide review
- lots of stuff on strings

PASS BY REFERENCE VS. PASS BY VALUE

PASS BY VALUE

Consider this relatively straightforward program. The function `sumThree` takes in three integers, then returns their sum as an integer:

```
1 #include <iostream>
2
3 using namespace std;
4
5 // returns the sum of three numbers
6 int sumThree(int a, int b, int c){
7     return a + b + c;
8 }
9
10 int main(){
11     int firstInt = 2;
12     int secondInt = 3;
13     int thirdInt = 5;
14
15     cout << "The sum of the numbers is: "
16         << sumThree(firstInt, secondInt, thirdInt) << endl;
17 }
18 }
```



one-line function that just returns a sum

PASS BY VALUE

When sumThree is called (on line 17), the *values* of the three variables given as arguments are copied into the parameters of sumThree

So when we hit line 17, we call sumThree. When we call sumThree, three new ints (named a, b, and c) are declared, and their values are set equal to the values of the variables given as arguments.

```
1 #include <iostream>
2
3 using namespace std;
4
5 // returns the sum of three numbers
6 int sumThree(int a, int b, int c){
7     return a + b + c;
8 }
9
10 int main(){
11     int firstInt = 2;
12     int secondInt = 3;
13     int thirdInt = 5;
14
15
16     cout << "The sum of the numbers is: "
17         << sumThree(firstInt, secondInt, thirdInt) << endl;
18 }
```

PASS BY VALUE

When `sumThree` is called (on line 19), the *values* of the three variables given as arguments are copied into the parameters of `sumThree`:

main:

`firstInt`



2

`secondInt`



3

`thirdInt`



5

`sumThree:`

`a`



2

`b`



3

`c`



5

`a`, `b`, and `c` are totally new variables, that are set equal to the values of `firstInt`, `secondInt`, and `thirdInt` when the `sumThree` function is called.

PASS BY REFERENCE

Now consider this version of the program:

it works exactly the same... but we pass in each of the variables *by reference*. This means that instead of *copying the value* stored in a particular variable, we *give the address of that variable*.

What's the difference?

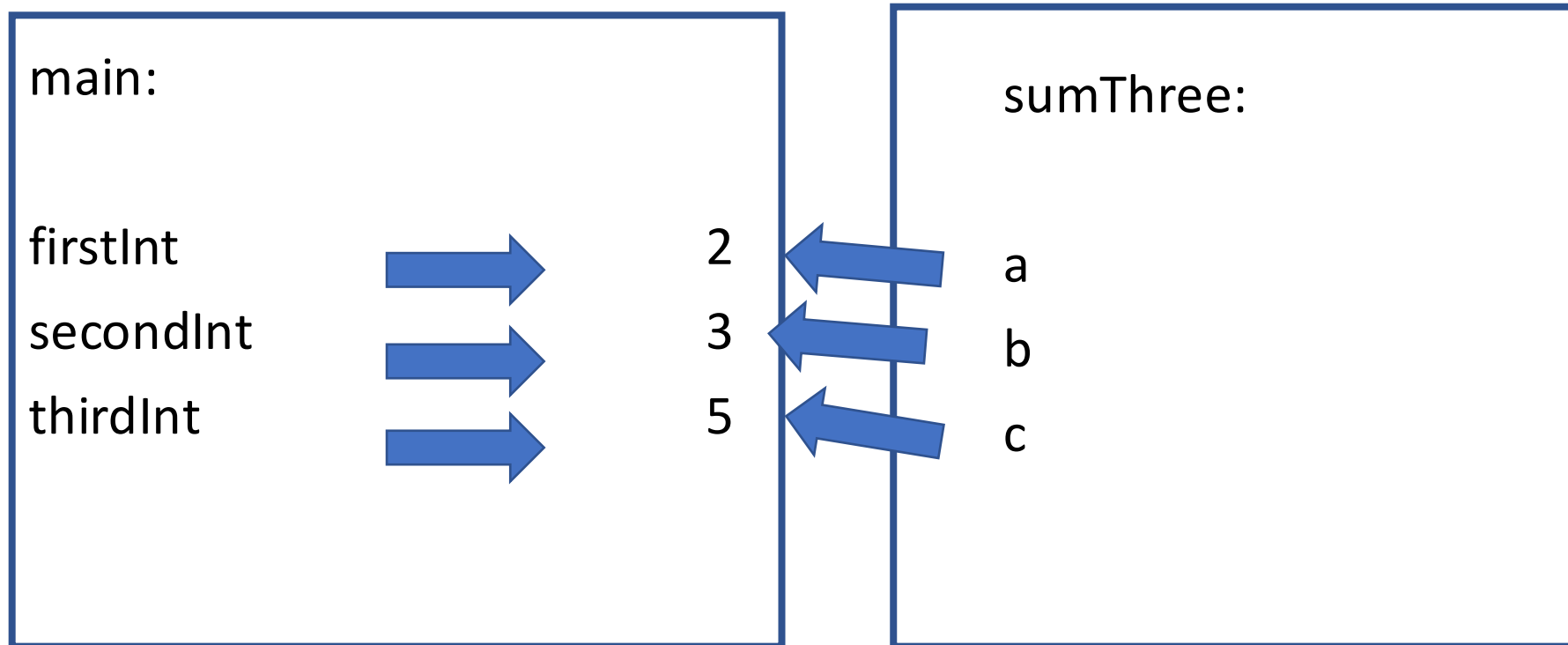
You indicate that a variable is passed by reference by putting a & sign before the variable name when specifying the parameters:



```
10 int sumThreeByReference(int &a, int &b, int &c){
11     return a+b+c;
12 }
13
14
15 int main(){
16     int firstInt = 2;
17     int secondInt = 3;
18     int thirdInt = 5;
19
20
21     cout << "The sum of the numbers is: "
22         << sumThreeByReference(firstInt, secondInt, thirdInt) << endl;
23 }
```

PASS BY REFERENCE

When `sumThreeByReference` is called on line 22, this time no new variables are made. 'a' in `sumThreeByReference` is *literally the same variable* as `firstInt` in `main`:



a, b, and c are not copies of `firstInt`, `secondInt`, and `thirdInt`. They are instead new names referring to `firstInt`, `secondInt`, and `thirdInt`.

PASS BY VALUE

What difference does it make? Let's try changing the values of variables in sumThree:

main:

firstInt



2

secondInt



3

thirdInt



5

sumThree:

a



2

b



3

c



5

a, b, and c are totally new variables, that are set equal to the values of firstInt, secondInt, and thirdInt when the sumThree function is called.

PASS BY VALUE

What difference does it make? Let's try changing the values of variables in sumThree:

```
5 // returns the sum of three numbers
6 int sumThree(int a, int b, int c){
7     a += 1;
8     return a + b + c;
9 }
10
11 // returns the sum of three numbers. all parameters passed by reference
12 int sumThreeByReference(int &a, int &b, int &c){
13     return a+b+c;
14 }
15
16
17 int main(){
18     int firstInt = 2;
19     int secondInt = 3;
20     int thirdInt = 5;
21
22     sumThree(firstInt, secondInt, thirdInt);
23
24     cout << "firstInt after SumThree: " << firstInt << endl;
25
26     sumThreeByReference(firstInt, secondInt, thirdInt);
27
28     cout << "firstInt after SumThreeByReference: " << firstInt << endl;
```

result: as expected, firstInt is unchanged



```
Bens-MacBook-Air-2:code-samples benallen$ ./sumThree
firstInt after SumThree: 2
```

PASS BY REFERENCE

What difference does it make? Let's try changing the values of variables in sumThreeByReference:

```
11 // returns the sum of three numbers. all parameters passed by reference
12 int sumThreeByReference(int &a, int &b, int &c){
13     a += 1;
14     return a+b+c;
15 }
16
17
18 int main(){
19     int firstInt = 2;
20     int secondInt = 3;
21     int thirdInt = 5;
22
23     sumThree(firstInt, secondInt, thirdInt);
24
25     cout << "firstInt after SumThree: " << firstInt << endl;
26
27     sumThreeByReference(firstInt, secondInt, thirdInt);
28
29     cout << "firstInt after SumThreeByReference: " << firstInt << endl;
```

result: after
sumThreeByReference, firstInt
is increased by 1

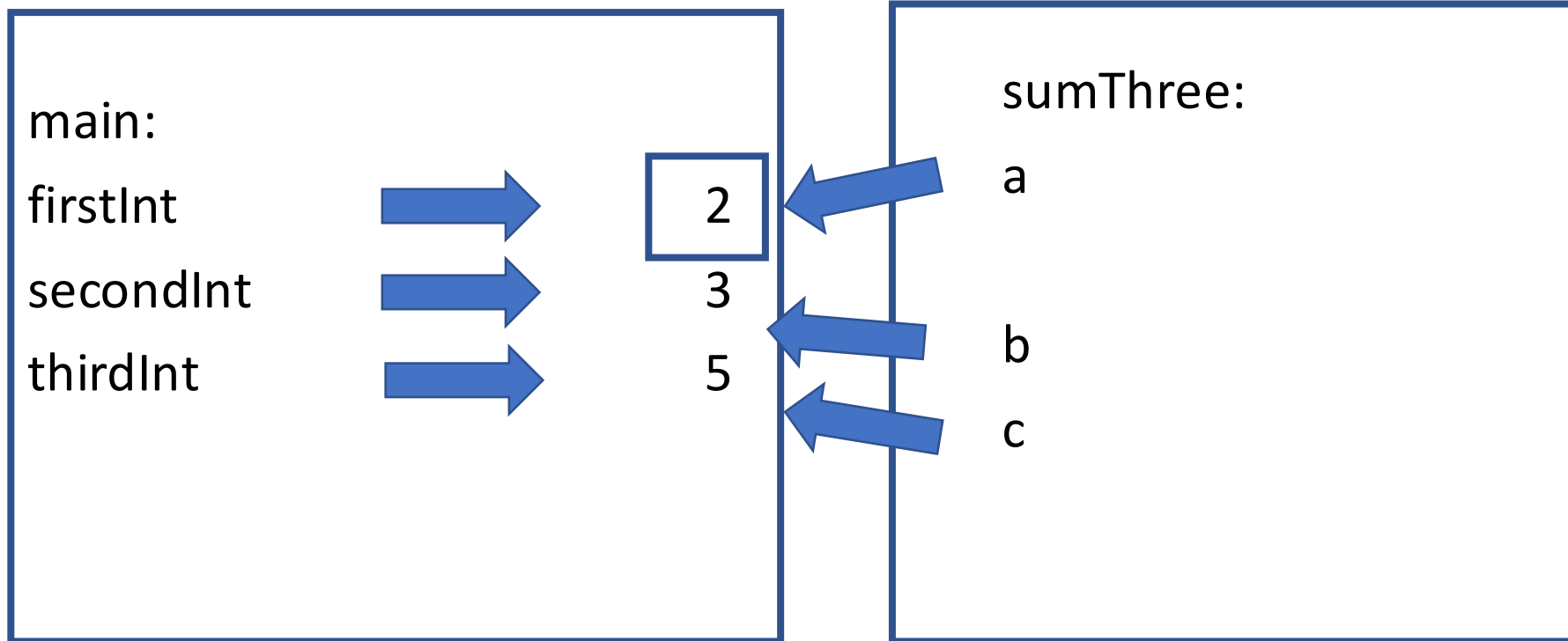


```
Bens-MacBook-Air-2:code-samples allen$ ./sumThree
firstInt after SumThree: 2
firstInt after SumThreeByReference: 3
Bens-MacBook-Air-2:code-samples allen$
```

PASS BY REFERENCE

This is because `firstInt` and `a` are *literally the same variable*.

Both `firstInt` and `a` are exactly the same thing, just by different names



`a`, `b`, and `c` are not copies of `firstInt`, `secondInt`, and `thirdInt`. They are instead new names referring to `firstInt`, `secondInt`, and `thirdInt`.

WHY PASS BY REFERENCE MATTERS #1

- Arrays are *always* pass-by-reference. Remember that if you change an array in a function, it changes it *everywhere*

this produces the following output:

```
Bens-MacBook-Air-2:code-samples benallen$ ./arrayChanger
myArray before arrayChanger:
0 1 2 3 4 5 6 7 8 9
myArray after arrayChanger:
137 1 2 3 4 5 6 7 8 9
```

```
6
7 void arrayChanger(int theArray[], int arrayLength){
8     if (arrayLength > 0){
9         theArray[0] = 137;
10    }
11 }
12
13 int main(){
14     int myArray[LENGTH];
15     for (int index = 0; index < LENGTH; index++){
16         myArray[index] = index;
17     }
18     cout << "myArray before arrayChanger: " << endl;
19     for (int index = 0; index < LENGTH; index++){
20         cout << myArray[index] << " ";
21     }
22     cout << endl;
23
24     arrayChanger(myArray, LENGTH);
25
26     cout << "myArray after arrayChanger: " << endl;
27     for (int index = 0; index < LENGTH; index++){
28         cout << myArray[index] << " ";
29     }
30     cout << endl;
31 }
```

STRING STUFF

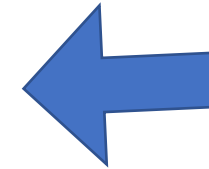
SOME STRING FUNCTIONS

Expression	Effect
<code>strVar.at(index)</code>	Returns the element at the position specified by <code>index</code> .
<code>strVar[index]</code>	Returns the element at the position specified by <code>index</code> .
<code>strVar.append(n, ch)</code>	Appends <code>n</code> copies of <code>ch</code> to <code>strVar</code> , where <code>ch</code> is a <code>char</code> variable or a <code>char</code> constant.
<code>strVar.append(str)</code>	Appends <code>str</code> to <code>strVar</code> .
<code>strVar.clear()</code>	Deletes all the characters in <code>strVar</code> .
<code>strVar.compare(str)</code>	Returns 1 if <code>strVar > str</code> ; returns 0 if <code>strVar == str</code> ; returns -1 if <code>strVar < str</code> .
<code>strVar.empty()</code>	Returns <code>true</code> if <code>strVar</code> is empty; otherwise it returns <code>false</code> .
<code>strVar.erase()</code>	Deletes all the characters in <code>strVar</code> .
<code>strVar.erase(pos, n)</code>	Deletes <code>n</code> characters from <code>strVar</code> starting at position <code>pos</code> .

apologies for
how ugly /boring these
slides are...

SOME STRING FUNCTIONS

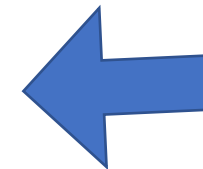
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note that these two are equivalent: the array subscript notation is included to maintain compatibility with C

SOME STRING FUNCTIONS

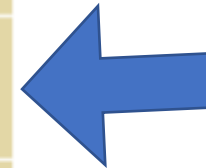
Expression	Effect
<code>strVar.at(index)</code>	Returns the element at the position specified by <code>index</code> .
<code>strVar[index]</code>	Returns the element at the position specified by <code>index</code> .
<code>strVar.append(n, ch)</code>	Appends <code>n</code> copies of <code>ch</code> to <code>strVar</code> , where <code>ch</code> is a <code>char</code> variable or a <code>char</code> constant.
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<code>strVar.erase()</code>	Deletes all the characters in <code>strVar</code> .
<code>strVar.erase(pos, n)</code>	Deletes <code>n</code> characters from <code>strVar</code> starting at position <code>pos</code> .



you can place
either chars or
other strings at
the end of a
string

SOME STRING FUNCTIONS

Expression	Effect
<code>strVar.at(index)</code>	Returns the element at the position specified by <code>index</code> .
<code>strVar[index]</code>	Returns the element at the position specified by <code>index</code> .
<code>strVar.append(n, ch)</code>	Appends <code>n</code> copies of <code>ch</code> to <code>strVar</code> , where <code>ch</code> is a <code>char</code> variable or a <code>char</code> constant.
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<code>strVar.erase()</code>	Deletes all the characters in <code>strVar</code> .
<code>strVar.erase(pos, n)</code>	Deletes <code>n</code> characters from <code>strVar</code> starting at position <code>pos</code> .

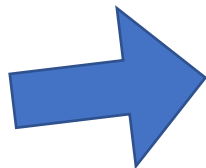


VERY USEFUL

MORE STRING FUNCTIONS

potential pitfall: technically, `str.length()` returns a variable of type `size_t`. it is best practice when printing a `size_t` to cast it to an `int`: i.e.,

```
cout << static_cast<unsigned int> strVar.length();
```



Expression	Effect
<code>strVar.find(str)</code>	Returns the index of the first occurrence of <code>str</code> in <code>strVar</code> . If <code>str</code> is not found, the special value <code>string::npos</code> is returned.
<code>strVar.find(str, pos);</code>	Returns the index of the first occurrence at or after <code>pos</code> where <code>str</code> is found in <code>strVar</code> .
<code>strVar.find_first_of(str, pos)</code>	Returns the index of the first occurrence of any character of <code>strVar</code> in <code>str</code> . The search starts at <code>pos</code> .
<code>strVar.find_first_not_of(str, pos)</code>	Returns the index of the first occurrence of any character of <code>str</code> not in <code>strVar</code> . The search starts at <code>pos</code> .
<code>strVar.insert(pos, n, ch);</code>	Inserts <code>n</code> occurrences of the character <code>ch</code> at index <code>pos</code> into <code>strVar</code> ; <code>pos</code> and <code>n</code> are of type <code>string::size_type</code> ; and <code>ch</code> is a character.
<code>strVar.insert(pos, str);</code>	Inserts all the characters of <code>str</code> at index <code>pos</code> into <code>strVar</code> .
<code>strVar.length()</code>	Returns a value of type <code>string::size_type</code> giving the number of characters in <code>strVar</code> .
<code>strVar.replace(pos, n, str);</code>	Starting at index <code>pos</code> , replaces the next <code>n</code> characters of <code>strVar</code> with all the characters of <code>str</code> . If <code>n > length</code> of <code>strVar</code> , then all the characters until the end of <code>strVar</code> are replaced.
<code>strVar.substr(pos, len)</code>	Returns a string which is a substring of <code>strVar</code> starting at <code>pos</code> . The length of the substring is at most <code>len</code> characters. If <code>len</code> is too large, it means "to the end" of the string in <code>strVar</code> .
<code>strVar.size()</code>	Returns a value of type <code>string::size_type</code> giving the number of characters in <code>strVar</code> .
<code>strVar.swap(str1);</code>	Swaps the contents of <code>strVar</code> and <code>str1</code> . <code>str1</code> is a <code>string</code> variable.

(for anyone who's had to work with regular c-strings (arrays of chars), you'll really appreciate every last one of these functions.)

CIN >> VS. GETLINE()

- `cin >>` reads a single whitespace-delimited token — it stops at the first space, tab, or newline, and leaves that whitespace sitting in the input buffer.
- `getline(cin, str)` reads an entire line — spaces and all — up to the next newline, then discards that newline.
- Rule of thumb: use `cin >>` for single numbers or single words; use `getline` for full lines of text, like names or sentences.

```
int age;  
cin >> age;  
// stops at the space/newline  
  
string name;  
getline(cin, name);  
// keeps internal spaces
```

“Jane Doe” typed after `cin >> age`: `cin >>` only grabs “Jane”; `getline` grabs the whole name.

THE CLASSIC BUG: MIXING CIN >> AND GETLINE

Calling `cin >> age` and then `getline(cin, name)` right after looks fine — but the second call silently reads an empty line. Here's why, and the fix.

BROKEN

```
int age;
cin >> age;

string name;
getline(cin, name);
// name is empty!

// the leftover \n from
// age's Enter key is what
// getline just read
```

FIXED

```
int age;
cin >> age;
cin.ignore();

string name;
getline(cin, name);
// name reads correctly

// cin.ignore() discards
// that leftover \n first
```

After any `cin >>`, an Enter key's newline is still waiting in the buffer — `cin.ignore()` clears it before the next `getline`.

WHY READ NUMBERS AS STRINGS?

- `cin >> num` reads a number directly, but bad input (letters, blank lines, extra spaces) puts `cin` into a fail state — every read after that silently fails too.
- Reading the whole line as a string first, then converting it yourself, lets you check the input and give a clear error before you ever try to convert it.
- The pattern: `getline(cin, line)` to grab the text, then `stoi` / `stod` to turn it into a number.

```
string line;  
cin >> line;  
// user typed "abc"  
  
int n = stoi(line);  
// throws an exception –  
// crashes if not caught
```

Reading into a string first means the program controls what happens next — the input itself can never break.

STRING TO NUMBER: STOI AND STOD

- `stoi(str)` converts a string to an int. `stod(str)` converts a string to a double — use `stod` whenever the value can have a decimal component.
- Both stop at the first character that doesn't fit the number and ignore anything after it, and both throw an exception if the string has no valid number at all.

READING AN INT

```
string line;
cout << "Age: ";
getline(cin, line);

int age = stoi(line);

cout << "Next year you'll "
     << "be " << age + 1;
```

READING A DECIMAL

```
string line;
cout << "Price: ";
getline(cin, line);

double price = stod(line);

cout << "With tax: "
     << price * 1.08;
```